JUNQI LU

GitHub | Google Scholar | Personal Website
Date of Birth: August 24, 2002 | Email: lu.junqi.001@student.uni.lu
Luxembourg

EDUCATION

University of Luxembourg Belval, Luxembourg

- **M.Sc. in Mathematics** Sep. 2026 – Jun. 2028 (Expected)
  - Mathematical Modelling and Computational Sciences track

Beijing Institute of Technology Beijing, China

- **B.S. in Computer Science** (Second Bachelor's Degree) Sep. 2024 – Jun. 2026
- **B.S. in Mathematics and Applied Mathematics** Oct. 2020 – Jun. 2024

PUBLICATIONS

(: Equal contribution; #: Corresponding author)*

[1] **Junqi Lu***, Ruixiang Sun*, Xin Li#, Gaopeng Peng, Ao Zhang, and Mingzhong Wang. Prevailing Bisimulation Metric Learning Is Biased: Implicit Regularization and Its Remedy. In *Advances in Neural Information Processing Systems* (**NeurIPS**), 2026. (**Accepted**)

[2] **Junqi Lu**, Bosen Liu, Cuicui Pei, Qingan Qiu#, and Li Yang. Learning to optimize termination decisions under hybrid uncertainty of system lifetime and task duration. *Computers & Industrial Engineering*, 206, 111208, 2025. DOI: 10.1016/j.cie.2025.111208 (**Published**)

[3] **Junqi Lu**, Qingan Qiu#, Rongchi Sun, and Yike Zhang. Simplicity Wins: How Myopic Decisions Enable Safe Learning in Mission-Critical Systems. (**Under Review**)

RESEARCH EXPERIENCE

Deep Reinforcement Learning & State Representation Sep. 2025 – Jun. 2026
Research Assistant | Advisor: Prof. Xin Li | 1029 DRL Lab | Beijing Institute of Technology

- **Theory:** Identified double-sampling bias in deep bisimulation metric learning; showed via Itô SDE analysis how transition noise induces implicit Jacobian regularization and representation collapse.
- **Method:** Developed BSPO, a Fenchel-dual saddle-point formulation with an auxiliary network that estimates expected Bellman residuals using single-transition samples.
- **Experiments:** Evaluated BSPO on stochastic/visually distracting DeepMind Control tasks and six Atari games; the work led to a **NeurIPS 2026 co-first-author paper**.

Reliability Engineering & Stochastic Control Jun. 2023 – Jun. 2026
Research Assistant | Advisor: Prof. Qingan Qiu | Beijing Institute of Technology

- **Model:** Formulated Bayesian MDPs for sequential task termination under uncertain system lifetimes and task durations, with Gamma–exponential updates from censored observations.
- **Theory:** Proved existence and monotonicity of optimal termination thresholds and structural monotonicity of value functions.
- **Algorithm:** Designed an $O(N)$ monotone threshold-update algorithm; numerical experiments showed up to 45.11% lower cost than a non-termination policy. First-author paper in *Computers & Industrial Engineering* (2025).

SELECTED PROJECTS

Nitrogen Cycling Dynamics Model (MCM-ICM 2025 Finalist) [GitHub: 100+ Stars]

- **Modeling**: Developed a dynamic ecosystem model using modified **Lotka-Volterra equations** to simulate nitrogen flow across forest and agricultural food webs.
- **Open Science**: Released a fully reproducible solution repository with comprehensive documentation and tutorials. Garnered **100+ GitHub Stars**, widely recognized by the modeling community for its code quality and educational value.
- **Award**: Selected as a **Finalist Winner (Top 2%)** out of 27,456 teams globally.

Formal Verification of a Mahjong Winning-Hand Checker [Demo] [Code]
AI4Math Summer School, Zhejiang University Summer 2026

- **Independent Project**: Conceived and completed the project individually, from problem selection and specification design to implementation, formal proofs, and the final summer-school presentation.
- **Formal Modeling**: Encoded three-suit Mahjong in **Lean 4**, separating declarative winning-hand specifications from an executable checker for standard hands and seven pairs, including declared melds and kongs.
- **Correctness Proofs**: Proved the executable checker sound and complete with respect to the declarative specification for every hand state: `isWinning state = true` ↔ `WinningSpec state`, extending validation beyond individual test cases.

HONORS & AWARDS

- **Meritorious Winner**, Mathematical Contest in Modeling (MCM) 2023, 2026
- **Outstanding Student Award** (Top 10% university-wide), BIT Nov. 2025
- **Academic Excellence Scholarship** (Rank 1st in CS 2nd Bachelor Program), BIT Oct. 2025
- **Finalist (Top 2%)**, Interdisciplinary Contest in Modeling (ICM) Feb. 2025
- **University "Excellent Thesis" Prize**, BIT Jun. 2024
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM) Nov. 2022

SKILLS

Research	Reinforcement Learning, Representation Learning, Bayesian MDPs, Stochastic Control, Mathematical Optimization, SDE Analysis
Programming	Python (PyTorch, NumPy, Pandas), C++, SQL, MATLAB, Bash
Tools	Linux, Git, Docker, LaTeX, Make, Lean 4 (Formal Verification)
Languages	Chinese (Native), English (B2), German (A1)

OUTREACH & INTERESTS

Technical Writing: CNBlogs — 20+ posts on algorithms and programming, 22K+ views.
Outside Research: Distance running (half-marathon PB: 1:41:34); guitarist and bassist in university rock bands.